

PRIORITIES

Step 1: Correlation

- Suppose we suspect: Obesity causes depression
- First step: Check if they occur together in data.
If obese people show more depression $\rightarrow$ correlation exists.
- But: Correlation $\neq$ causation
- Because: 1) depression might cause obesity. 2) both might be caused by something else (eg: low income)
- This is called a common cause problem.

Step 2: Controlling variables

- To test if income is the real cause: Compare people with same income level.
- If obesity — depression link disappears $\rightarrow$ income was the hidden cause. This is called: "controlling for a variable"

Step 3: Controlled experiments

Best method: Change only one factor and hold others constant.

Example: Temperature affects insect size.

- raise insects at different temperatures
- keep everything else same

wk	March 2024							wk	April 2024						
	Mo	Tu	We	Th	Fr	Sa	Su		Mo	Tu	We	Th	Fr	Sa	Su
09								14	1	2	3	4	5	6	7
10	4	5	6	7	8	9	10	15	8	9	10	11	12	13	14
11	11	12	13	14	15	16	17	16	15	16	17	18	19	20	21
12	18	19	20	21	22	23	24	17	22	23	24	25	26	27	28
13	25	26	27	28	29	30	31	18	29	30					